Explore-exploit trade-off in vocal learning

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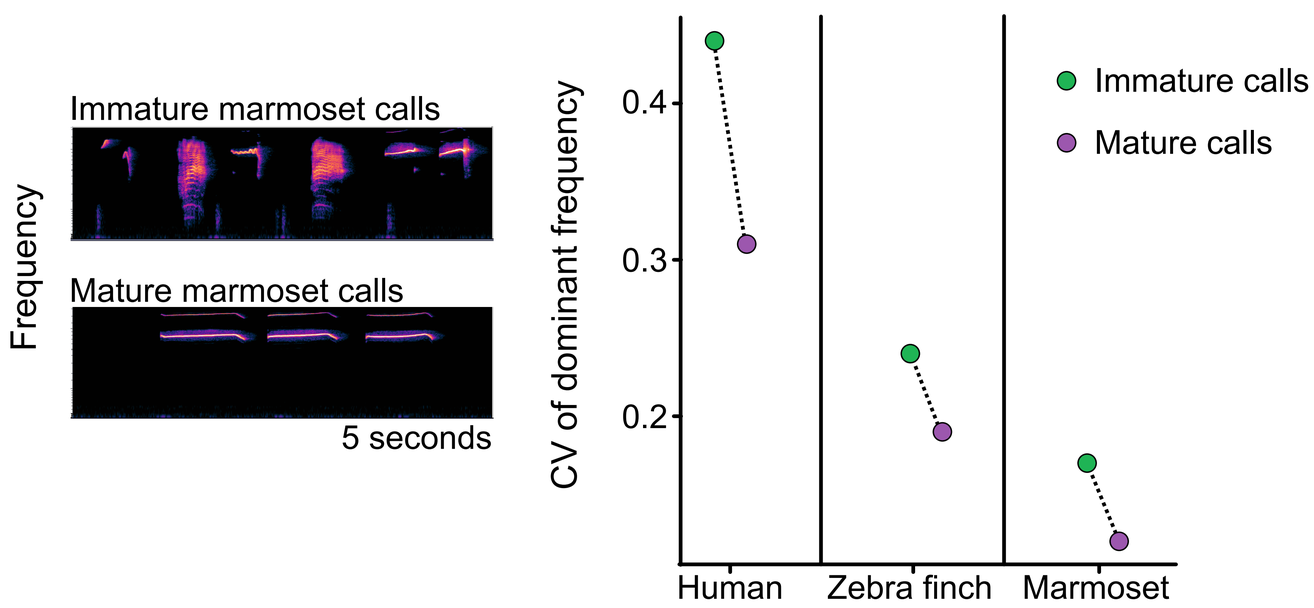
Early **exploration** facilitates vocal development

- What is the role of **imitation** and of **exploration** in vocal learning?

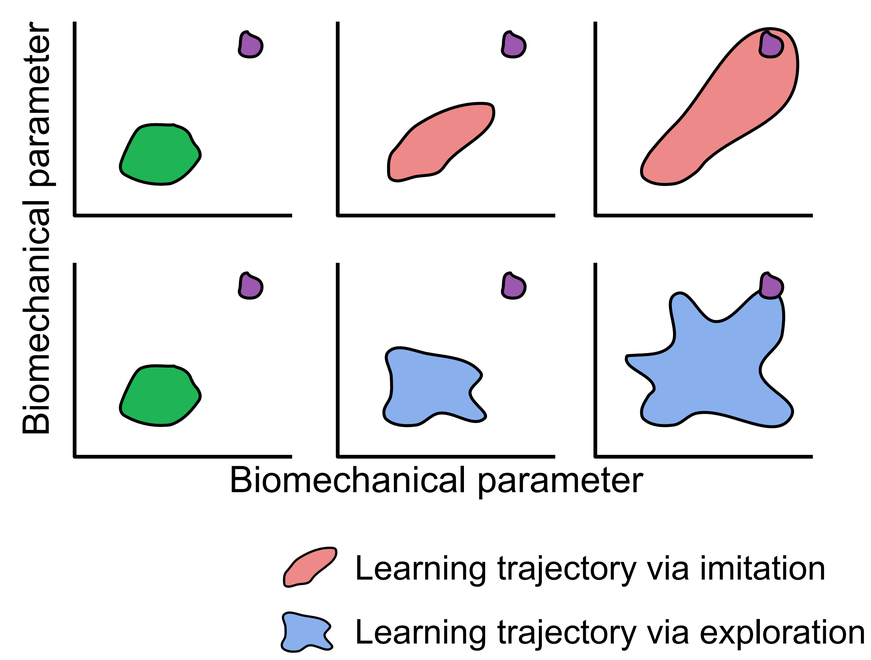
- How to model this process in an **embodied** manner?

- How do infants solve the problem of "**poverty of stimulus**" when learning?

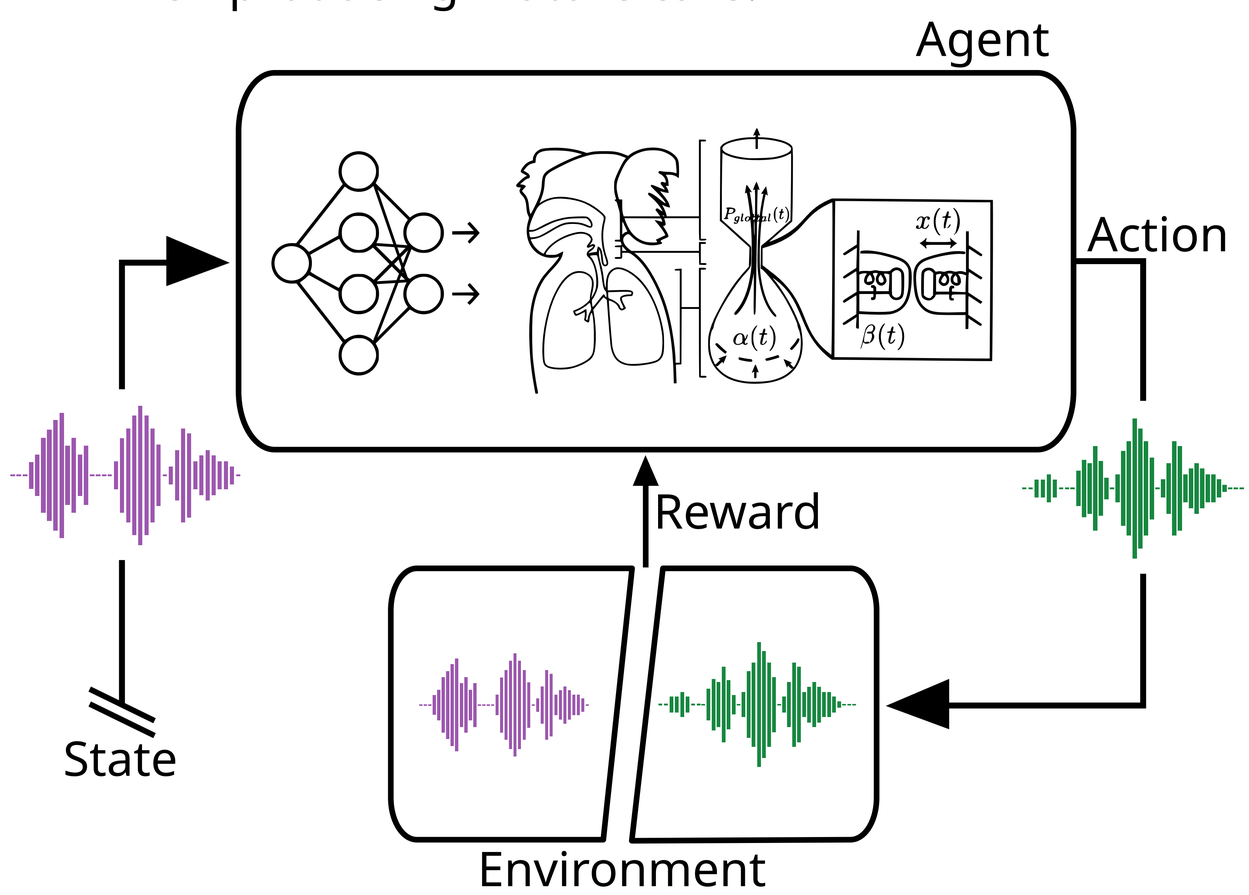
1 Immature vocalizations have **higher variability** than mature vocalizations in vocal learning animals.



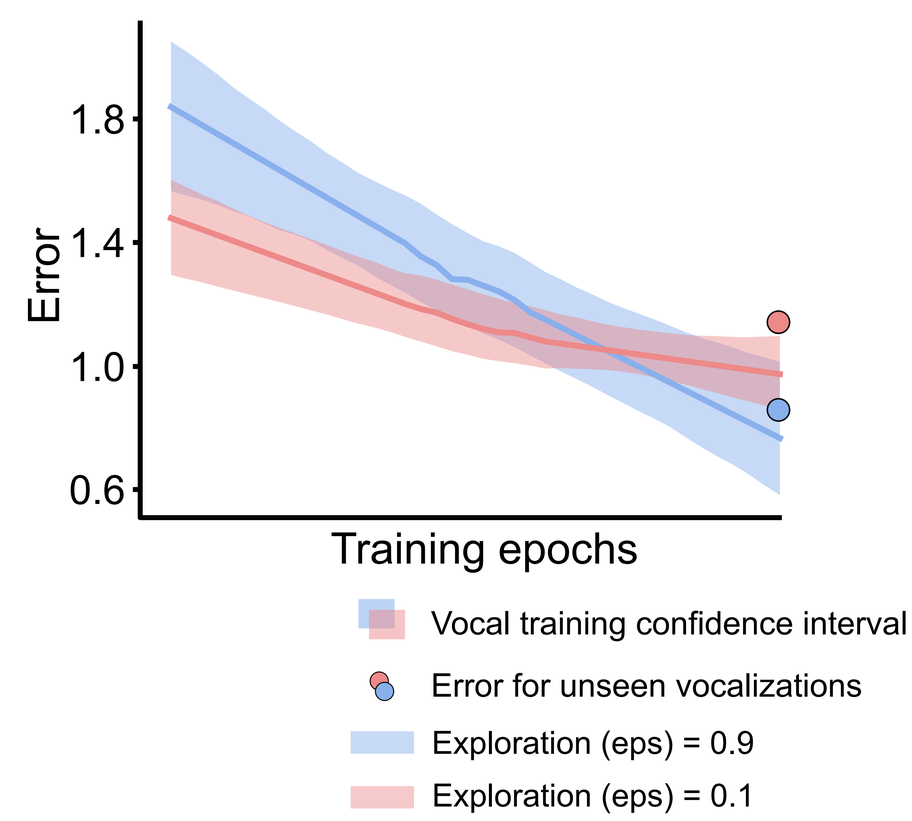
2 Schematics of the space of biomechanical parameters and learned vocalizations over time.



3 RL agent is a biomechanical model of vocal tract, and receives reward when producing mature calls.



4 The learning process is **faster and more robust** when there's higher exploration.



References:
Lipkind D., Geambasu A., Levelt, CC., *ToCS* (2019);
Warlaumont AS., Westerman G., Buder EH., Oller DK.. *NN* (2013).

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Funding:
NSF Graduate Fellowship DGE-2039656 (TTV);
NIH-NINDS R01NS054898 (AAG).

Acknowledgements:
Yisi Zhang,
Renata Biazzi